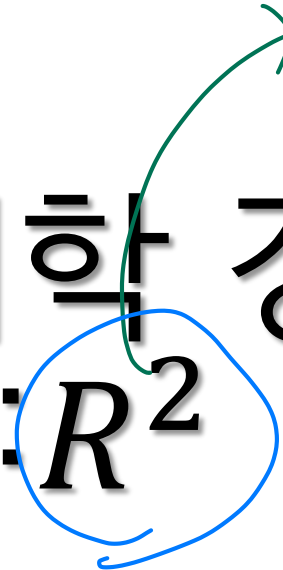


Goodness of fit

계량경제학 강의 10
: R^2



1. R^2 (R – square)

Command window: `genr e_hat=resid`

Equation: UNTITLED Workfile: UNTITLED::Untitled\

Dependent Variable: INCOME
Method: Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/15/21 Time: 18:18
Sample: 1 32
Included observations: 32
INCOME=C(1)+C(2)*EDUCATION+C(3)*GENDER

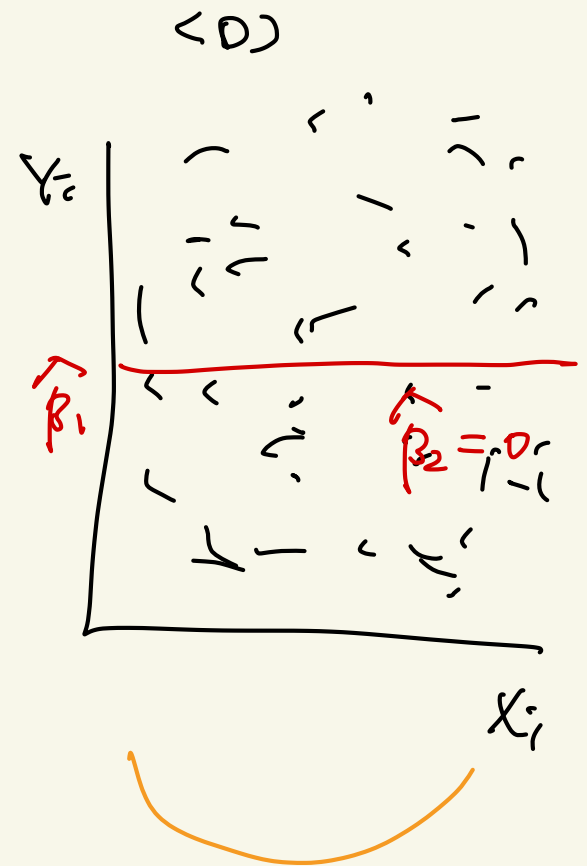
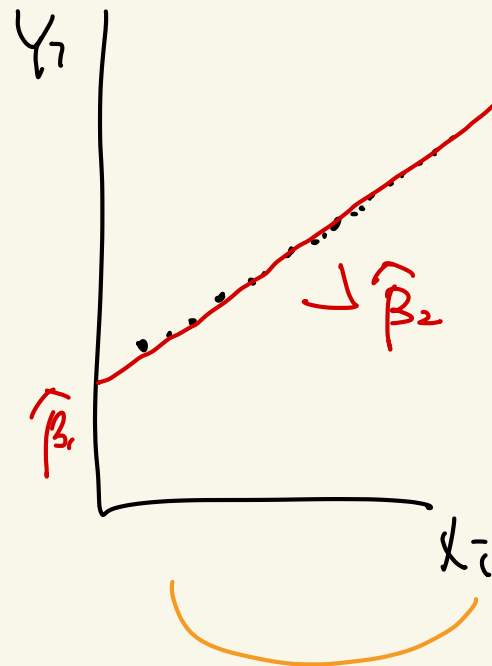
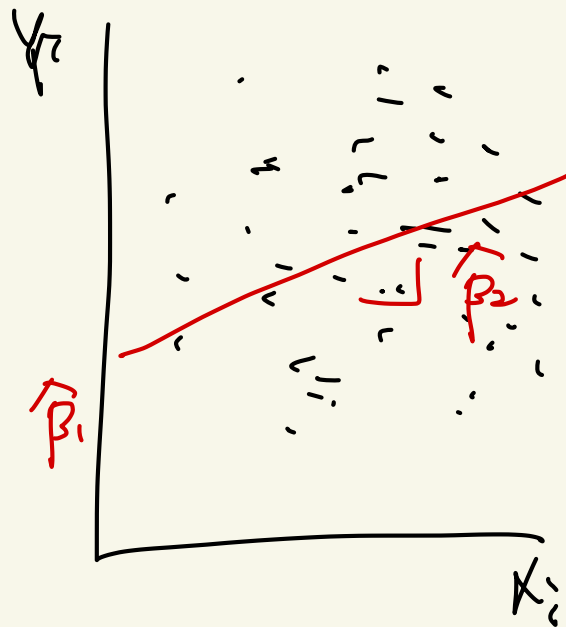
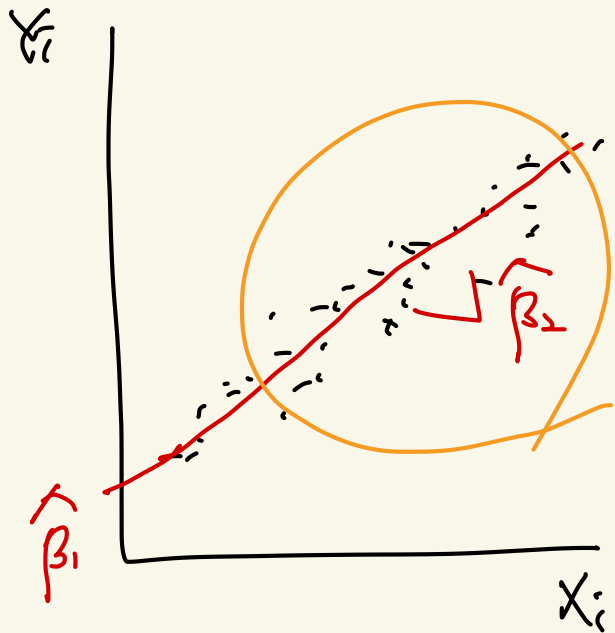
	Coefficient	Std. Error	t-Statistic	Prob.
C(1)	17.80774	5.509647	3.232102	0.0031
C(2)	2.585757	0.314338	8.226030	0.0000
C(3)	-9.667076	2.720810	-3.553014	0.0013

R-squared	0.743388	Mean dependent var
Adjusted R-squared	0.725690	S.D. dependent var
S.E. of regression	7.622096	Akaike info criterion
Sum squared resid	1684.794	Schwarz criterion
Log likelihood	-108.8246	Hannan-Quinn criterion
F-statistic	42.00551	Durbin-Watson stat
Prob(F-statistic)	0.000000	

all coef=.

$$\text{Q1)} \quad Y_i = \beta_1 + \beta_2 X_{2i} + e_i$$

<A> R^2 > R^2 <C>



$$0 < R^2 < 1$$

$$R^2 = 1$$

$$R^2 = 0$$

$\Rightarrow \text{case C} > \text{case A} > \text{case B} > \text{case D}$

coefficient of determination, 결정계수

1. R^2 (R - square)

○ X_1 의 변화가 Y_2 의 변화
를 설명하는 정도라 적당 R^2

- 표본회귀방정식

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{e}_i$$

이 식을 deviation from mean으로 고쳐쓰면

$$\Rightarrow (\overset{y_i}{Y_i} - \bar{Y}) = \hat{\beta}_2 (\overset{x_i}{X_{2i}} - \bar{X}) + \hat{e}_i$$

양변을 제곱하여 더하면
 $\Rightarrow y_i = \hat{y}_i + \hat{e}_i$
 $\hat{\beta}_2 x_i$

$$(y_i)^2 = (\hat{\beta}_2 x_i + \hat{e}_i)^2$$

$$\begin{aligned} y_i^2 &= \hat{\beta}_2^2 x_i^2 + \hat{e}_i^2 + 2 \cdot \hat{\beta}_2 x_i \hat{e}_i \\ \sum y_i^2 &= \underbrace{\sum \hat{\beta}_2^2 x_i^2}_{\sum \hat{y}_i^2} + \sum \hat{e}_i^2 + 2 \cdot \hat{\beta}_2 \underbrace{\sum x_i \hat{e}_i}_{=0} \quad \left(\begin{array}{l} \because \text{by f.o.c} \\ \sum x_i \hat{e}_i = 0 \end{array} \right) \end{aligned}$$

1. R^2 (R - square)

$$\sum_{i=1}^n y_i^2 = \sum_{i=1}^n \hat{y}_i^2 + \sum_{i=1}^n \hat{e}_i^2$$

양변을 n 으로 나누면

$$\begin{array}{ccc} \text{T.S.S} & \text{ESS} & \text{RSS} \\ \frac{1}{n} \sum y_i^2 & = & \frac{1}{n} \sum \hat{y}_i^2 + \frac{1}{n} \sum \hat{e}_i^2 \\ \text{total sum of } y_i & \text{Explained sum of } y_i & \text{Residual sum of } y_i \end{array}$$

$\frac{\sum_{i=1}^n y_i^2}{n}$ 은 y 의 분산의 추정치라고 볼 수 있다.

1. R^2 (R - square)

$$\frac{\sum_{i=1}^n y_i^2}{n} = \frac{\sum_{i=1}^n \hat{y}_i^2}{n} + \frac{\sum_{i=1}^n \hat{e}_i^2}{n}$$

Total sum of squares = explained sum of squares + residual sum of squares

$$\text{TSS} = \text{ESS} + \text{RSS}$$

"Definition"

(그래프)

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

• 만약, X_i 가 Y 와 $\approx$ 관계가 강해지면 $\uparrow$, $\underline{\underline{R^2 \uparrow}}$

$R^2 \uparrow$ 것이 항상 best? No!

2. adjusted R^2

- R^2 의 한계: 설명변수를 추가할 수록 커짐

예) $Y_i = \hat{\beta}_1^* + \hat{\beta}_2^* X_{2i} + \hat{e}_i^*$ — ① $\Rightarrow R^2 = 0.8$

$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \hat{e}_i$ — ② $R^2 = 0.805$ ✓

불필요한.

X_{3i} 가 불필요한 변수임에도 불구하고

$\downarrow$
 $\rightarrow \sum \hat{y}_i^2 \uparrow$
 $\Leftrightarrow \sum \hat{e}_i^2 \downarrow$
 $\Rightarrow \textcircled{2} \quad R^2 \uparrow$

$$\underbrace{\sum_{i=1}^n \hat{e}_i^{*2}} \geq \underbrace{\sum_{i=1}^n \hat{e}_i^2}$$

2. adjusted R^2

penalty 부여.

- adjusted R^2 : 설명변수를 많이 포함시킨 것에 대해 일정한 패널티를 부여

$$R^2 = 1 - \frac{\sum \widehat{e}_i^2}{\sum y_i^2}$$

$$\rightarrow \overline{R^2} = 1 - \frac{\sum \widehat{e}_i^2 / (n-k)}{\sum y_i^2 / (n-1)}$$

$\Rightarrow$ 추가된 설명변수
효율성 $\downarrow$
불순도 $\uparrow$

"Principle of parsimony"

$\overline{R^2}$ 의 원리.

① $k \uparrow$ 복잡도

$\rightarrow$ ① $\overline{R^2} \downarrow$ 작아짐 ($\because$ penalty)

$\rightarrow$ ② $R^2 \uparrow$ (model fit 이 더

④ 증명한 변수가 추가되었으면?

$R^2 \uparrow$ (④의 효과가
효율적이거나 아닐까?)

⑤ 불필요한 변수?
($\overline{R^2} \downarrow$)

3. R^2 와 F-test

- 모집단 회귀식 : $Y_i = \beta_1 + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + e_i$
- 귀무가설 : $H_0: \beta_2 = \beta_3 = \cdots = \beta_k = 0$
- Unrestricted regression : $Y_i = \widehat{\beta}_1 + \widehat{\beta}_2 X_{2i} + \cdots + \widehat{\beta}_k X_{ki} + \widehat{e}_i$
- Restricted regression : $Y_i = \widehat{\beta}_1^* + \widehat{e}_i^*$

3. R^2 와 F-test

- $f - value = \frac{(\sum_{i=1}^n \hat{e}_i^{*2} - \sum_{i=1}^n \hat{e}_i^2)/(k-1)}{\sum_{i=1}^n \hat{e}_i^2/(n-k)} \sim F(k-1, n-k)$